

# ISyE 6669 HW 15

Fall 2025

**Note:** There may be more than one correct way to model constraints in this homework. Verify that the formulation models the logical relations correctly and are linear (with integer variables).

1. Let  $x_1, x_2, x_3$  be continuous nonnegative variables with values in the range  $[0, 10]$ . Write a set of constraints to model the requirement that:

$$|2x_1 - x_2 - x_3| \geq 2$$

by introducing an additional binary variable.

**Solution:** Since  $|2x_1 - x_2 - x_3| \geq 2$  is equivalent to:

$$2x_1 - x_2 - x_3 \leq -2 \quad \text{OR} \quad 2x_1 - x_2 - x_3 \geq 2$$

we can model it using following constraints:

$$\begin{aligned} 2x_1 - x_2 - x_3 &\geq 2y - M(1 - y) \\ 2x_1 - x_2 - x_3 &\leq -2(1 - y) + My \\ y &\in \{0, 1\} \end{aligned}$$

2. Let  $x_1, x_2, x_3$  be binary variables with the following relationship:

| $x_1$ | $x_2$ | $x_3$ |
|-------|-------|-------|
| 0     | 0     | 0     |
| 0     | 1     | 1     |
| 1     | 0     | 1     |
| 1     | 1     | 0     |

Table 1: Truth table for XOR operation

- (a) In the following formulation correct? Explain what each constraint is doing?

$$\begin{aligned} (1 - x_1) + x_2 + x_3 &\geq \frac{1}{2} \\ x_1 + (1 - x_2) + x_3 &\geq \frac{1}{2} \\ x_1 + x_2 + (1 - x_3) &\geq \frac{1}{2} \\ (1 - x_1) + (1 - x_2) + (1 - x_3) &\geq \frac{1}{2} \end{aligned}$$

| $x_1$ | $x_2$ | $x_3$ |
|-------|-------|-------|
| 0     | 0     | 1     |
| 0     | 1     | 0     |
| 1     | 0     | 0     |
| 1     | 1     | 1     |

Table 2: Variable combinations not allowed under XOR

**Solution:** We are modeling  $x_3 = x_1 \vee x_2$ , where  $\vee$  represents XOR relation. Formulation is correct because it allows all possibilities from Table ??, and it is excluding all other possibilities, which are listed in Table 2.

First three constraints are ensuring that whenever one variable is equal to 1, at least one more variable is also equal to 1. Fourth constraint is ensuring that all the variables are not equal to 1 at the same time.

- (b) Can you write a formulation using one equation? Hint: You may need to use an additional variable.

**Solution:** Yes,  $x_3 = x_1 \vee x_2$  can be modeled using single constraint:

$$x_1 + x_2 = x_3 + 2z$$

where  $z$  is binary variable.

3. A plantation is in the business of commercially growing and harvesting trees for lumber. The plantation is divided into 10 grids as shown below:

|   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|----|

We are planning for the next four years. Each year, the plantation cuts all the trees in some of the grids and then replants trees in these grids in the same year. Use the following decision variable:

$$x_{it} = \begin{cases} 1 & \text{if trees in grid } i \text{ are harvested in year } t \\ 0 & \text{otherwise} \end{cases}$$

- (a) A profit of  $p_{it}$  is earned if the grid  $i$  is harvested in time period  $t$ . Write down a linear objective function that represents maximizing total profit.

**Solution:**

$$\max \sum_{i=1}^{10} \sum_{t=1}^4 p_{it} x_{it}$$

- (b) No more than 6 grids can be harvested in a single year. Write linear inequalities to model this constraint.

**Solution:**

$$\sum_{i=1}^{10} x_{it} \leq 6, \forall t = 1, 2, 3, 4$$

- (c) Because trees need time to grow, each grid may be harvested at most 2 times over the next four years, and it may not be harvested in consecutive years. Write linear inequalities to model these constraints.

**Solution:** Constraints

$$x_{it} + x_{i,t+1} \leq 1, \forall i = 1, \dots, 10, \forall t = 1, 2, 3$$

will ensure that same grid is not harvested in consecutive years and consequently each grid will be harvested at most 2 times.

- (d) Ecological considerations require the implementation of following rule: In any given year, the set of unharvested grids must be connected. For example: harvesting grids 1, 2, 5, 6 in time period 1 is not allowed since the unharvested grids are disconnected into two blocks 3, 4 and 7, 8, 9, 10. Write linear inequalities to represent the above constraints. You may introduce artificial variables as necessary.

**Solution:** To ensure that each year unharvested grids are connected, we need to ensure that in each year there is at most one starting unharvested grid i.e. there is at most one "switch" from 1 to 0 between consecutive grids. We can do this by introducing binary variables  $z_{it}$ ,  $i = 1, \dots, 10$ ,  $t = 1, 2, 3, 4$  which will indicate if grid  $i$  is unharvested grid coming right after harvested grid, and using constraints:

$$\begin{aligned} z_{1t} &\geq 1 - x_{1t}, \forall t = 1, 2, 3, 4 \\ z_{it} &\geq x_{i-1,t} - x_{it}, \forall i = 2, \dots, 10, \forall t = 1, 2, 3, 4 \\ \sum_{i=1}^{10} z_{it} &\leq 1, \forall t = 1, 2, 3, 4 \end{aligned}$$

First line of constraints will indicate whether unharvested block starts at first grid or not, second line of the constraints will indicate whether unharvested block starts at any of the grids 2, ..., 10 and finally last constraints will enforce that in each year we only have at most one starting grid of the unharvested block.

4. A hospital is scheduling nurses for aiding in surgery. 10 surgeries need to be performed. Each surgery requires exactly 2 nurses. There are 25 available nurses. We have the following additional constraints.
- (a) A nurse can aid in a surgery only if the nurse is qualified to do so. The following data is available:  $d_{ij} = 1$  if nurse  $i$  is qualified to aid in surgery  $j$  and  $d_{ij} = 0$  if nurse  $i$  is not qualified to aid in surgery  $j$  ( $d_{ij}$  provided for all  $i = 1, \dots, 25$  and for all  $j = 1, \dots, 10$ ).
  - (b) Nurse 1, Nurse 2, ..., Nurse 15 are senior nurses. Nurses 16, ..., Nurse 25 are junior nurses. Each surgery must have at least 1 senior qualified nurse.
  - (c) Nurse 1 and Nurse 20 do not work well together. They must not be scheduled to the same surgery.
  - (d) Each nurse can aid in at most 2 surgeries.
  - (e) At least 5 junior nurses must be aiding surgeries.

Formulate an integer program to find a feasible schedule.

**Solution:** Let  $x_{ij}$ ,  $i = 1, \dots, 25$ ,  $j = 1, \dots, 10$  represents binary variables indicating whether nurse  $i$  is assigned to surgery  $j$ , and let  $y_i$ ,  $i = 16, \dots, 25$  represents binary variables indicating whether junior nurse  $i$  is assigned to any of the surgeries. Then we can formulate problem as:

$$\min \quad 1 \tag{1}$$

$$\text{s.t. } x_{ij} \leq d_{ij}, \quad i = 1, \dots, 25, j = 1, \dots, 10 \tag{2}$$

$$x_{1j} + x_{20j} \leq 1, \quad j = 1, \dots, 10 \tag{3}$$

$$\sum_{i=1}^{25} x_{ij} = 2, \quad j = 1, \dots, 10 \tag{4}$$

$$\sum_{i=1}^{15} x_{ij} \geq 1, \quad j = 1, \dots, 10 \tag{5}$$

$$\sum_{j=1}^{10} x_{ij} \leq 2, \quad i = 1, \dots, 25 \tag{6}$$

$$\sum_{j=1}^{10} x_{ij} \geq y_i, \quad i = 16, \dots, 25 \tag{7}$$

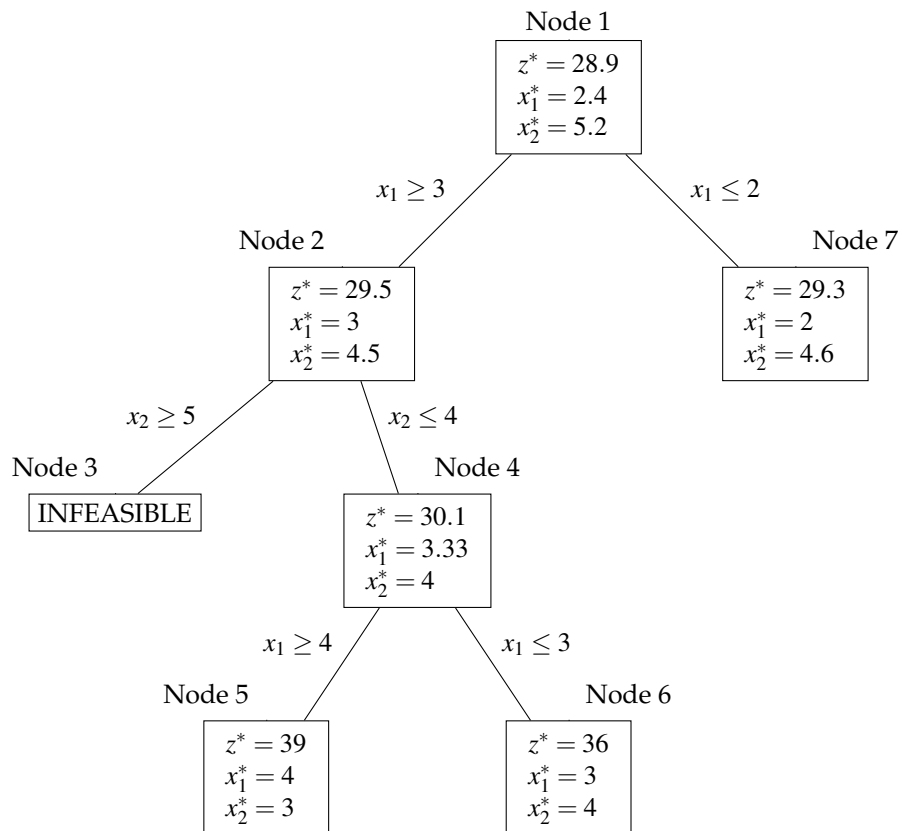
$$\sum_{i=16}^{25} y_i \geq 5 \tag{8}$$

$$x_{ij} \in \{0, 1\}, \quad i = 1, \dots, 25, j = 1, \dots, 10 \tag{9}$$

$$y_i \in \{0, 1\}, \quad i = 16, \dots, 25. \tag{10}$$

where constraints (4) are ensuring that every surgery has exactly two nurses assigned, constraints (2) are ensuring that nurse assigned to a surgery is qualified for that surgery, constraints (5) are ensuring that every surgery has at least one senior nurse assigned, constraints (3) are ensuring that Nurse 1 and Nurse 20 do not get assigned to the same surgery, constraints 6 are ensuring that each nurse is assigned to at most two surgeries and finally constraints 7 and 8 are ensuring that at least five junior nurses get assignments.

5. Consider the following branch-and-bound tree for solving a two variable integer programming problem.



(a) Is this a minimization or a maximization problem?

**Solution:** Problem is minimization.

(b) For each of the following nodes, indicate whether you need to further branch on the node.

Node 3:

Node 5:

Node 7:

**Solution:**

Node 3: No, because problem is infeasible.

Node 5: No, because problem has integer solution.

Node 7: Yes, because problem has fractional solution with objective value which is better than the current upper bound.

(c) According to this branch-and-bound tree, what is the range (lower and upper bounds) in which the optimal IP value lies? Explain your answer.

**Solution:**

Lower bound: 29.3.

Upper bound: 36.